# CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
## Third Semester of M.Sc. (Mathematics) Examination November 2019
## MA813 Topology-2

Date: 11/11/2019, Monday Time: 01:30 p.m. to 02:15 p.m. Maximum Marks: 20

**Multiple Choice Questions**

**Important Instructions:**

1. Tick the correct answer and it should be written in question paper itself.
2. You can use only non-programmable calculator. No other gadget is allowed in the examination hall.
3. Each question carries one mark.
4. **Notations** $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$ and $\mathbb{R}$ denote the set of natural numbers, integers, rational numbers and real numbers respectively. Other notations are standard.

**Q - I** Choose the correct answer from the given options in the followings: [20]

**1.** The set $\mathbb{N}$ with discrete topology is______ space.

(a) a $T_1$ (c) not a $T_1$

(b) not a $T_2$ (d) none of these

**2.** If $\tau_d$ and $\tau_u$ represent the discrete and usual topologies on $\mathbb{R}$ respectively and $I_{\mathbb{R}}$ denotes the identity map, then the map $I_{\mathbb{R}} : (\mathbb{R}, \tau_d) \to (\mathbb{R}, \tau_u)$ is ______.

(a) discontinuous (c) continuous

(b) one one & onto (d) none of these

**3.** Let $X$ be a non-empty set and $\mathcal{A}$ be the collection of subsets of $X$ such that the union of members of the set $\mathcal{A}$ is X then $\mathcal{A}$ is ______.

(a) not a basis (c) not a topology

(b) a subbasis (d) none of these

**4.** If $X = \mathbb{R}_\ell$, $Y = [-1, 1]$, then ______ is not an open set in $Y$.

(a) $\left(\frac{1}{2}, 1\right)$ (c) $\left[\frac{1}{2}, 1\right)$

(b) $\left(\frac{1}{2}, 1\right]$ (d) none of these

**5.** For $a, b \in \mathbb{Q}$, the interval $(a, b)$ is homeomorphic to ______, with respect to the usual topology.

(a) $\mathbb{Q}$ (b) $[a, b)$ (c) $\{a, b\}$ (d) $\mathbb{R}$

**6.** Consider $\mathbb{R}$ with usual topology. Then the set $\mathbb{Q}$ as a subspace of is ______.

(a) closed but not open (c) neither closed nor open

(b) clopen (d) none of these

**7.** If $X = \mathbb{R}$, $Y = (-\infty, 1) \cup [2, \infty)$ of $\mathbb{R}$, then $Y$ is ______.

(a) open w.r.t. $\mathbb{R}_\ell$ (c) closed w.r.t. $\mathbb{R}_\ell$

(b) closed w.r.t. $\mathbb{R}_u$ (d) none of these

**8.** With respect to usual topology on $\mathbb{R}$, if $Y = \left\{\frac{1}{n} : n \in \mathbb{N}\right\}$, then '0' is ______ of $Y$.

(a) a cluster point (c) not a cluster point

(b) an interior point (d) none of these

**9.** For $X = \{0, 1, 2\}$, $\mathcal{S} = \{\{0, 1\}, \{0, 2\}\}$ is not a subbasis with respect to the topology ______.

(a) $\{\phi, \{0, 1\}, \{0, 2\}, \{0, 1, 2\}\}$

(b) $\{\phi, \{1\}, \{0, 1\}, \{0, 2\}, \{0, 1, 2\}\}$

(c) $\{\phi, \{0, 1, 2\}\}$

(d) none of these

**10.** The set $X = \{1, 2, \dots, 2019\}$ with the usual topology is ______.

(a) not a $T_2$ space

(b) not a compact space

(c) not a $T_1$ space

(d) a $T_2$ space

**11.** For $i \in \mathbb{N}$, consider $(X_i, \tau_i)$ as a topological space. The basis element for the product topology on the set $X = \prod_{i=1}^{\infty} X_i$ are of the form $B = \prod_{i=1}^{\infty} B_i$, where ______.

(a) for $i = 1, 2, \dots, n,\ B_i = U_i \in \tau_i$ and $B_j = X_j$ for $j \neq i$

(b) for all $i,\ U_i \in \tau_i$

(c) for $\{i_1, i_2, \dots i_n\} \subset \mathbb{N},\ B_{i_j} \in \tau_{i_j}$ but $B_k = X_k$ for $k \neq \{i_1, i_2, \dots i_n\}$

(d) none of these

**12.** The element of the form $\left\{\frac{a}{b}\right\},\ a, b \in \mathbb{Q},\ b \neq 0$ as a subspace of $\mathbb{R}_u$ is ______.

(a) open (b) not open (c) closed (d) None of these

**13.** If $S^n$ denote the closed unit sphere in $\mathbb{R}^{n+1}$, then one point compactification of $\mathbb{R}^n$ is homeomorphic to ______.

(a) $S^n$

(b) $S^n - p,\ p = (0, 0, \dots, 0, 1) \in \mathbb{R}^{n+1}$

(c) $S^{n+1}$

(d) None of these

**14.** Consider the space $X = [2017, 2018] \cup [2019, 2020]$ as a subspace of $\mathbb{R}_u$. Then ______.

(a) $[2017, 2018]$ and $[2019, 2020]$ are not closed sets

(b) $[2017, 2018]$ and $[2019, 2020]$ are not open sets

(c) $[2017, 2018]$ and $[2019, 2020]$ are clopen sets

(d) None of these

**15.** The sets $\{(x,y) \in \mathbb{R}^2 : x^2+y^2=1\} \cup \{(x,y) \in \mathbb{R}^2 : x^2+y^2=4x-3\}$ are ______ with respect to usual topology of $\mathbb{R}^2$.

(a) not connected
(b) connected
(c) totally disconnected
(d) None of these

**16.** The set $(-\infty, 0) \cup [1, \infty)$ is ______ in $\mathbb{R}_\ell$.

(a) regular
(b) connected
(c) not totally disconnected
(d) none of these

**17.** Urysohn lemma gives a rich supply of continuous real valued functions from a ______.

(a) $2^{nd}$ countable and Hausdorff space
(b) $2^{nd}$ countable space
(c) regular space
(d) $2^{nd}$ countable and regular space

**18.** Suppose $\{X_\alpha : \alpha \in \Lambda\}$ is a collection of locally compact spaces. Then $\prod\limits_{\alpha \in \Lambda} X_\alpha$ is locally compact if ______.

(a) each $X_\alpha$ is compact for finitely many but for all

(b) each $X_\alpha$ is compact for finitely many

(c) each $X_\alpha$ is compact for all but finitely many

(d) None of these

**19.** Arbitrary product of connected spaces need not be connected with respect to ______ topology.

(a) product
(b) box
(c) usual
(d) None of these

**20.** The set $(0,1)$ has a compactification given by the map $f(x) =$ ______, $x \in \mathbb{R}$.

(a) $\frac{1}{x}$
(b) $\sin x$
(c) $\cos x$
(d) $(\cos 2\pi x, \sin 2\pi x) \in \mathbb{R}^2$

Candidate's Seat No.________________

**CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY**
**Third Semester of M.Sc. (Mathematics) Examination November 2019**
**MA813 Topology-2**

Date: 11/11/2019, Monday    Time:02:15 p.m. to 04:30 p.m.    Maximum Marks:50

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**Instructions:**

1. **Section I and Section II must be written in separate answer books.**
2. You can use only non-programmable calculator. No other gadget is allowed in the examination hall.
3. Figures to the right indicate marks.
4. **Notations:** $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$ and $\mathbb{R}$ denote the set of natural numbers, integers, rational numbers and real numbers respectively. Other notations are standard.

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## SECTION I

**Q - II** (A) Attempt any TWO of the following. [**8**]

(1) Let $X$ and $Y$ be spaces and $X \times Y$ be the space with product topology. If $\mathfrak{B}$ and $\mathfrak{C}$ be the bases for the topologies $X$ and $Y$ respectively then prove that $\mathfrak{D} = \{B \times C : B \in \mathfrak{B},\ C \in \mathfrak{C}\}$ is a basis for a topology on $X \times Y$.

(2) Prove that a second countable regular space is a normal space.

(3) By drawing the Topologist's Sine curve, explain that it is connected but not locally connected space.

(B) Attempt any SIX of the following. [**12**]

(1) Define first and second countable space.

(2) State and prove the Pasting lemma.

(3) Prove that every metrizable space is a Hausdorff space.

(4) Prove that every normal space is completely regular.

(5) Is the statement " Union of two connected space is also connected." true or false? Justify.

(6) Prove that every component in a topological space is closed.

(7) State real Stone-Weierestrass theorem.

(8) Define normal and completely regular spaces.

## SECTION II

**Q - III** (A) Attempt any THREE of the following. [**18**]

(1) Let $X$ and $Y$ be topological spaces and $f : X \to Y$. Then prove that following are equivalent:

1. $f$ is continuous.
2. $f^{-1}(F)$ is closed in $X$ for each closed subset $F$ of $Y$.
3. $f(\overline{A}) \subseteq \overline{f(A)}$, for all $A \subseteq X$.
4. $\overline{f^{-1}(B)} \subseteq f^{-1}(\overline{B})$, for all $B \subseteq Y$.

(2) If $X$ and $Y$ are compact then prove that $X \times Y$ is compact with respect to product topology.

(3) Prove that $\mathbb{R}_\ell^2 = \mathbb{R}_\ell \times \mathbb{R}_\ell$ is not normal with respect to the product topology.

(4) **(a)** Let $X$ and $Y$ be topological spaces. Then Prove that $X \times Y$ is connected if and only if $X$ and $Y$ are connected.

**(b)** Prove that a space is locally connected if and only if every component of an open set is open.

(5) State Tietze extension theorem. Let $A$ be a closed subset of a normal space $X$. If $f : A \to [-r, r]$, $r \in \mathbb{R}^+$ is a continuous function, then prove that there exists a continuous function $g : X \to \left[\frac{-r}{3}, \frac{r}{3}\right]$ such that $|g(a) - f(a)| \leq \dfrac{2}{3}r$, for all $a \in A$.

(B) Attempt any TWO of the following. [**12**]

(1) **(a)** Let $\mathfrak{B}$ and $\mathfrak{B}'$ be bases for a topologies $\tau$ and $\tau'$ on $X$ respectively. Then prove that following are equivalent:

(i) $\tau'$ is finer than $\tau$.

(ii) For each $x \in X$ and each $B \in \mathfrak{B}$ containing $x$, there exists $B' \in \mathfrak{B}'$ such that $x \in B' \subseteq B$.

**(b)** Let $(X, \tau)$ be a topological space. Suppose $\mathfrak{C}$ is a collection of open subsets of $X$ such that for each $x \in X$ and each open set $U$ of $X$, there exists $C \in \mathfrak{C}$ such that $x \in C \subset U$, then prove that $\mathfrak{C}$ is a basis for a topology $\tau$ on $X$.

(2) Let $X$ be a locally compact Hausdorff space which is not compact and $Y$ be one point compactification of $X$. Then prove that $Y$ is Hausdorff space, $X$ is a subspace of $Y$, $Y - X$ is a single point and $\overline{X} = Y$.

(3) Let $X$ be a normal space and $P$ denote the set of all rational numbers in $[0, 1]$. For a neighborhood $U_p$ of $p \in P$, prove that whenever $p, q \in P$ and $p < q$, the re exists a neighborhood $U_q$ of $q$ such that $\overline{U_p} \subset U_q$.